Warm-up / A faster bus route

Find the missing bridge

Name: _____ Date: _____

Suggested lesson time: 20 minutes. Work individually, then compare when invited.

Useful words

Assumption: An idea the reasoning relies on without stating or establishing it.

Prediction: A claim about what will happen or what an untested case will show.

Relevant difference: A difference that could affect whether a result carries over.

Read the scenario

Invented scenario: An express service reduces journeys on Route A by ten minutes. A planner says, 'The same express arrangement will save ten minutes on Route B.' Route B has not been tested.

Produce: Write the observed result, one assumption, and one possible difference.

Your steps

1. State what was observed on Route A.

2. Name a feature that would need to be similar enough for the same time saving on Route B.

3. Describe a possible Route B difference that could reduce the saving.

Before you finish: Is your difference relevant to journey time, and clearly a possibility rather than a supplied fact?

Your response

Main activity / Will the checkout result carry over?

Name: _____

Read the scenario

Invented scenario: A store tests a checkout arrangement on Saturday and Sunday. Average waits fall from six minutes to four. The manager concludes, 'This arrangement will also reduce waits on weekdays.' No weekday test has been done.

Produce: Write the observed result, prediction, assumption, and one possible weakening difference.

Your steps

1. Copy the observed reason and the prediction separately.
2. Complete: 'The manager assumes that weekday conditions are similar enough in ___ for the weekend result to carry over.' Name a relevant feature such as customer volume or staffing.
3. Describe one possible weekday difference that would weaken that bridge. You are proposing a possibility, not claiming it happened.

Before you finish: Does your assumption connect this weekend result to weekdays, rather than merely say that faster checkout is desirable?

Your response

Exit ticket / A practice session for another class

Name: _____

Read the scenario

Invented scenario: A teacher gives her morning class ten minutes of vocabulary practice each day. Its average score rises from 60 to 75 on a weekly vocabulary quiz. She predicts the same routine will raise the evening class's score by 15 points. The evening class has not tried it.

Produce: Write the observed result and prediction, then an assumption and a possible weakening difference.

Your steps

1. Separate the morning result from the evening prediction.

2. Name one learning condition that would need to be similar enough for the prediction.

3. Describe a possible difference that could make the evening class improve less.

Before you finish: Does your assumption concern learning conditions, rather than just the desirability of higher scores?

Your response